

Disproof of conjecture 00000000211: the lucky primes contain no twin pair

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: The lucky numbers are obtained from the positive integers by the lucky sieve (at each round deleting the s_k -th position, where s_k is the k -th surviving number). Conjecture: The density of the intersection of lucky numbers and primes is asymptotically an explicit Euler-type constant, and the intersection contains infinitely many twin pairs; the constant is given by the first-order correction to independence of the two sieves. (lucky-prime intersection density) (The Chinese text states the same.)

Reading. The conjecture is a conjunction of (D) a density clause and

(T) the set $\mathcal{L} \cap \mathcal{P}$ of lucky primes contains infinitely many twin pairs,

where $\mathcal{L}$ is the set of lucky numbers and $\mathcal{P}$ the set of primes. A *twin pair* in $\mathcal{L} \cap \mathcal{P}$ is a pair $(p, p+2)$ with both members in $\mathcal{L} \cap \mathcal{P}$; this is the usual gap-two meaning (“a twin prime is a prime that has a prime gap of two”, Wikipedia, *Twin prime*). We prove that $\mathcal{L} \cap \mathcal{P}$ contains *no* twin pair, so (T) is false, and hence so is the conjunction (D) and (T), whatever (D) means. We do not formalize or address (D); in Lean it is an arbitrary proposition. Only the gap-two meaning of “twin” is refuted; pairs with other gaps (4, 6, ...) are not considered.

2 The lucky sieve

OEIS A000959 (“Lucky numbers”) describes the sieve as follows: “Start with the natural numbers. Delete every 2nd number, leaving 1 3 5 7 ...; the 2nd number remaining is 3, so delete every 3rd number, leaving 1 3 7 9 13 15 ...; now delete every 7th number, leaving 1 3 7 9 13 ...; now delete every 9th number; etc.” Its data begin 1, 3, 7, 9, 13, 15, 21, 25, 31, 33, ... Deletion is by *position* in the current list: the 1-indexed positions $m, 2m, 3m, \dots$ are deleted.

Definition 1 (Lean: `deleteEvery`, `stage`, `sievingNumber`, `isLucky`). Represent a list of survivors by its increasing enumeration $L : \mathbb{N} \rightarrow \mathbb{N}$ (0-indexed). For $m \geq 2$ put $(\text{del}_m L)(j) = L(j + \lfloor j/(m-1) \rfloor)$. Let $S_0(i) = i+1$ (the positive integers), $m_k = S_k(\max(1, k))$ and $S_{k+1} = \text{del}_{m_k} S_k$. A positive integer n is *lucky* if n lies in the range of S_k for every k , i.e. it is never deleted.

Lemma 2 (faithfulness). (a) (*range_deleteEvery*, *range_stage_succ*) For $m \geq 2$ and any L , the range of $\text{del}_m L$ is $\{L(i) : (i+1) \not\equiv 0 \pmod{m}\}$; so round $k+1$ deletes exactly the entries of S_k at 1-indexed positions divisible by m_k , and $m_k \geq 2$ (*two_le_sievingNumber*). Every S_k is strictly increasing (*stage_strictMono*). (b) (*sievingNumber_first_five*) $m_0, \dots, m_4 = 2, 3, 7, 9, 13$. (c) (*sievingNumber_eq_lucky*) For $k \geq 1$, $m_k = \ell_k$, where $\ell_j = S_{j+1}(j)$. (d) (*isLucky_iff*, *lucky_*

strictMono) n is lucky iff $n = \ell_j$ for some j , and ℓ is strictly increasing; so ℓ_j is the $(j + 1)$ -th lucky number. (e) (*lucky_first_ten*) $\ell_0, \dots, \ell_9 = 1, 3, 7, 9, 13, 15, 21, 25, 31, 33$, the first ten terms of A000959; there are infinitely many lucky numbers (*isLucky_infinite*).

Proof sketch. (a) Write $m = d + 1$. The map $j \mapsto j + \lfloor j/d \rfloor$ sends $j = dq + r$ ($0 \leq r < d$) to $mq + r$, which is a bijection onto $\{i : i \bmod m \neq m - 1\}$. (c),(d) use $S_k(i) \geq i + 1$, $S_{k+1}(i) \geq 2i + 1$ and the stability $S_{k+1}(j) = S_k(j)$ for $j < k$ (*stage_succ_of_lt*, *stage_stable*). (b),(e) are kernel *decide* computations. $\square$

Convention on indexing. The statement says the round deletes every s_k -th survivor, s_k the k -th surviving number. Read literally with $s_1 = 1$ this deletes everything, so we use the standard sieve of A000959: the first round uses 2 (the 2nd positive integer), and by Lemma 2(c) round $k \geq 2$ uses the k -th lucky number $s_k = \ell_{k-1}$ (3, 7, 9, 13, ...), exactly as the statement describes. The argument below uses only the first two rounds ($m_0 = 2$, $m_1 = 3$).

3 The proof

Lemma 3 (*stage_one*, *stage_two*, *stage_two_mod_six*). $S_1(j) = 2j + 1$ and $S_2(j) = 2(j + \lfloor j/2 \rfloor) + 1$. Hence every value of S_2 is $\equiv 1$ or $3 \pmod{6}$.

Proof. $m_0 = S_0(1) = 2$, so $S_1(j) = S_0(2j) = 2j + 1$. Then $m_1 = S_1(1) = 3$, so $S_2(j) = S_1(j + \lfloor j/2 \rfloor)$. For $j = 2q + r$, $r \in \{0, 1\}$, this is $6q + 2r + 1$. $\square$

Lemma 4 (*range_stage_anti*, *isLucky_mod_six*). The ranges of S_k decrease in k . Every lucky number is $\equiv 1$ or $3 \pmod{6}$. In particular 5 is not lucky (*not_isLucky_five*).

Proof. $S_{k+1}(j) = S_k(\cdot)$, so $\text{range } S_{k+1} \subseteq \text{range } S_k$. A lucky number lies in $\text{range } S_2$. $\square$

Theorem 5 (*eq_three_of_luckyPrime_of_prime_add_two*, *luckyPrimes_no_gap_two*, *twinPairs_eq_empty*). If p is a lucky prime and $p + 2$ is prime, then $p = 3$. No two lucky primes differ by 2, so the set of twin pairs in $\mathcal{L} \cap \mathcal{P}$ is empty.

Proof. If $p \equiv 1 \pmod{6}$ then $3 \mid p + 2$, so the prime $p + 2$ equals 3 and $p = 1$, which is not prime. If $p \equiv 3 \pmod{6}$ then $3 \mid p$, so $p = 3$. If p and $p + 2$ are both lucky primes, then $p = 3$, but $p + 2 = 5$ is not lucky. $\square$

Theorem 6 (*conjecture211_false*). For every proposition D : $\neg(D \wedge \text{the set of twin pairs in } \mathcal{L} \cap \mathcal{P} \text{ is infinite})$.

Also proved: *twin_lucky_primes_not_infinite* (the set of p with $p, p + 2$ both lucky primes is not infinite; it is empty) and *weak_twin_finite* (even requiring only the smaller member to be a lucky prime and the larger to be prime gives only $p = 3$).

4 Lean formalization and axiom audit

File `lean/Conjecture211/Basic.lean` (278 lines, namespace `Conjecture211`, only import `Mathlib`). Main statement:

```
def twinPairs : Set (N x N) :=
  {pq | pq.1 in luckyPrimes /\ pq.2 in luckyPrimes /\ pq.2 = pq.1 + 2}
theorem conjecture211_false (DensityClaim : Prop) :
  not (DensityClaim /\ twinPairs.Infinite)
```

(ASCII rendering; `luckyPrimes = {p | IsLucky p /\ p.Prime}`.) `lake build` succeeds.
`#print axioms conjecture211_false` gives `[propext, Classical.choice, Quot.sound]`;
`lucky_first_ten` depends on no axioms. There is no `sorry` and no `native_decide`.

Sources

- OEIS A000959, Lucky numbers, <https://oeis.org/A000959> (definition and data quoted above).
- Wikipedia, *Twin prime*, https://en.wikipedia.org/wiki/Twin_prime: “a twin prime is a prime that has a prime gap of two.”
- OEIS A031157, Numbers that are both lucky and prime, <https://oeis.org/A031157>, which remarks that “lucky numbers cannot be congruent to 2 (mod 3).” (Not used in the proof.)